Exponentials & Logarithms

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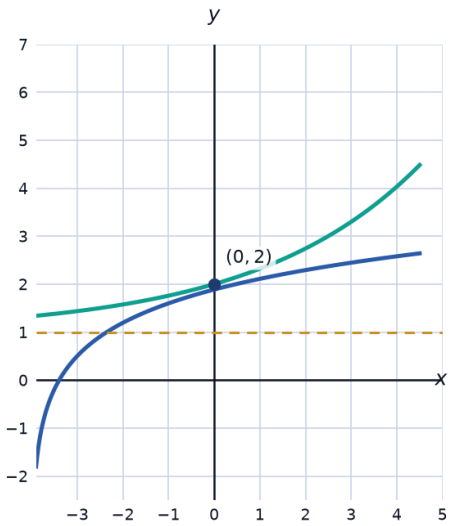
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Logarithm definition and evaluation

Rewrite $\log_3 81 = 4$ in exponential form, and verify.

2 [Foundational] • Target: Logarithm laws

Simplify $\log 8 + \log 5$ as a single logarithm.



3 [Foundational] • Target: Logarithm laws

Write $3\log_2 x + \log_2 y$ as a single logarithm.

4 [Foundational] • Target: Exponential growth and decay models

A population of bacteria starts at 300 and doubles every 4 hours. Write an exponential model and find the population after 12 hours.

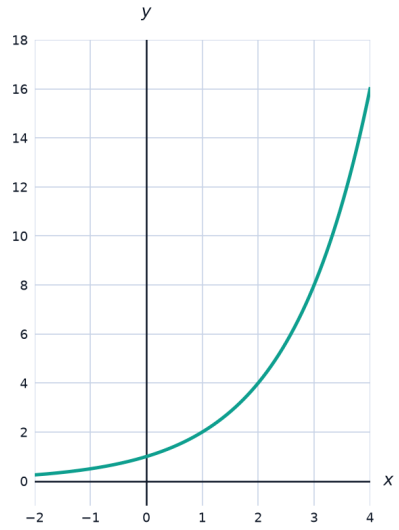
5 [Foundational] • Target: Exponential growth and decay models

A car worth 24{,}000 dollars depreciates at 12 percent per year. Find its value after 3 years, to the nearest dollar.

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6 [Foundational] • Target: Graphs of exponential functions

The graph of $y = 2^x$ is shown below. Use the graph to estimate the value of x for which $y = 8$, then confirm your answer algebraically.



7 [Foundational] • Target: Natural logarithms and e

Solve $e^x = 20$ for x, to the nearest hundredth, using natural logarithms.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: A well-designed word problem: comparing two investment accounts

This question has three parts. An investment of 5000 dollars is placed in an account that grows according to $A(t) = 5000e^{0.04t}$, where t is measured in years. (a) Find the value of the investment after 10 years, to the nearest dollar. (b) Find how many years it takes for the investment to double, to the nearest tenth of a year, using natural logarithms. (c) A second account offers simple (non-compounded) growth of 5 percent per year on the same initial 5000 dollars. Determine which account has the greater value after 10 years, and by how much, to the nearest dollar.

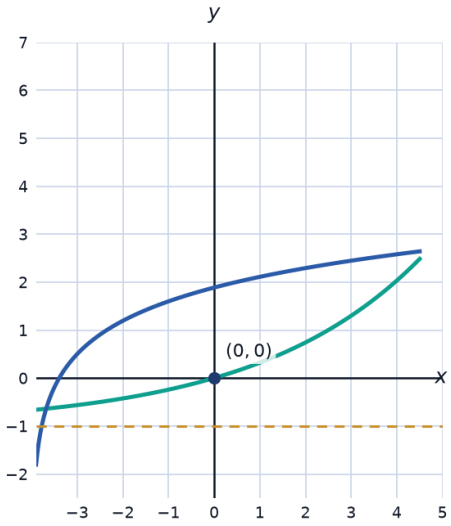
9 [Intermediate] • Target: A well-designed word problem: comparing two investment accounts

Solve the equation $5e^{0.04t} = 45$ for t in terms of natural logarithms, and find its value rounded to two decimal places.

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10 [Intermediate] • Target: Exponent rules

Solve the logarithmic equation $\log_3(x^2 - 7) = 2$ for all real solutions, and verify that all roots lie within the valid domain.



11 [Intermediate] • Target: Solving exponential equations

Solve the exponential equation $3^{2x-1} = 5^{x+2}$ for x in terms of natural logarithms.

12 [Intermediate] • Target: Logarithm definition and evaluation

Simplify the expression $\left(\frac{16x^8y^{-4}}{81z^{12}}\right)^{-3/4}$ and write your answer using only positive exponents.

13 [Intermediate] • Target: Logarithm laws

A population of bacteria in a petri dish starts with 250 organisms and quadruples every 5 hours. How many hours will it take for the population to reach 16,000?

14 [Intermediate] • Target: Solving logarithmic equations

Two assembly line components operate independently. Component A has a 95% reliability rate (probability of functioning properly is 0.95), and Component B has a 90% reliability rate (probability of functioning is 0.90). What is the probability that at least one of the two components fails?

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

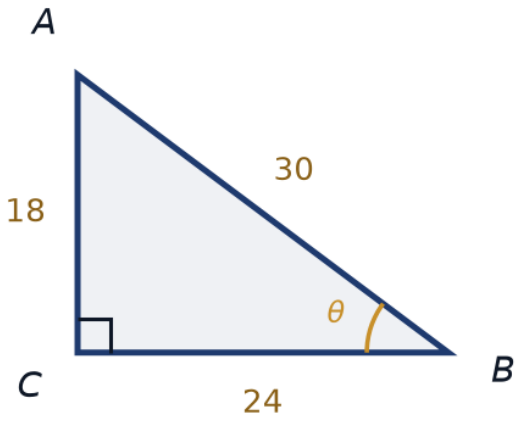
15 [Challenging] • Target: Exponential growth and decay models • Cross-Lesson: 1.01 Heart of Algebra

An investment fund gains 30% in value during Year 1, decreases by 20% in Year 2, and increases by 10% in Year 3. Determine the overall net percent change of the fund across the three-year period.

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16 [Challenging] • Target: Graphs of exponential functions • Cross-Lesson: 2.01 Quadratics & Polynomials

Two angles of a triangle are 55° and 75°. Find the third.



17 [Challenging] • Target: Natural logarithms and e • Cross-Lesson: 3.01 Functions & Graphs

A solid metal cone with base radius 4 cm and height 9 cm is submerged into a cylindrical beaker with internal radius 6 cm partially filled with water. Assuming the cone is fully submerged and no water overflows, by how many centimeters does the water level in the beaker rise?

18 [Challenging] • Target: A well-designed word problem: comparing two investment accounts • Cross-Lesson: 4.01 Ratios, Rates & Data

Consider the ordered dataset of 12 numbers: 14, 16, 18, 20, 22, 24, 26, 28, 30, 34, 38, 72. (a) Determine Q_1, the median, Q_3, and the IQR. (b) Use Tukey's $1.5 \times \text{IQR}$ rule to determine whether 72 is an outlier.

19 [Challenging] • Target: Exponent rules • Cross-Lesson: 5.01 Statistics & Probability

An equilateral triangle has a side length of 12 cm. Find: (a) the exact length of its altitude, and (b) its exact area.

20 [Challenging] • Target: Solving exponential equations • Cross-Lesson: 6.01 Geometry

A rectangular water storage reservoir has dimensions 2.5 meters by 3 meters by 1.2 meters. A drainage pump empties the reservoir at a constant rate of 75 liters per minute. Given that $1\ \text{m}^3 = 1000$ liters, how many hours will it take to completely empty the full reservoir?

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Exponent rules

A population of bacteria starts at 488 and doubles every 6 hours. Find the population after 12 hours.

a. 2440

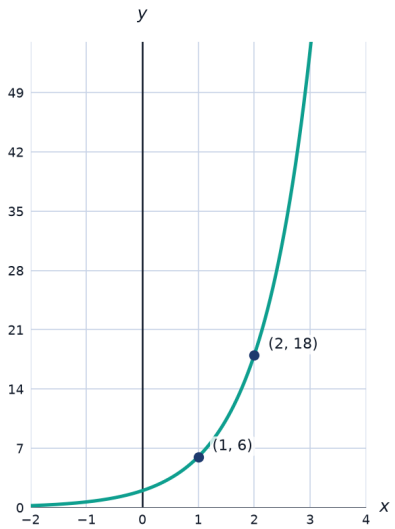
b. 1952

c. 976

d. 2928

22 [Foundational] • Target: Solving exponential equations

The graph of the exponential function $f(x) = a \cdot b^x$ passes through the points $(1, 6)$ and $(2, 18)$. Determine the value of a.



a. 2

b. 3

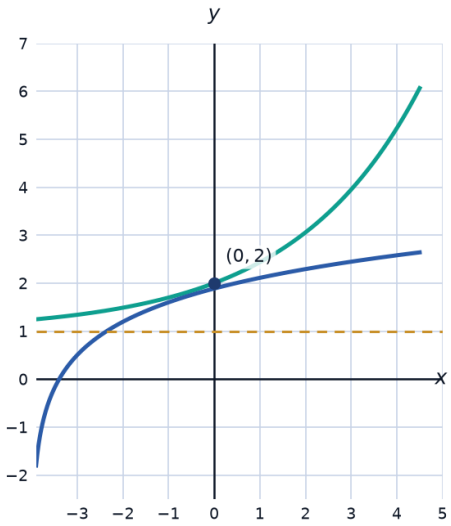
c. 6

d. 9

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23 [Foundational] • Target: Logarithm definition and evaluation

The graph of the exponential function $f(x) = a^x$ passes through the point $(2, 9)$. What is the value of a?



a. 3

b. 9

c. $\sqrt{3}$

d. 81

24 [Foundational] • Target: Logarithm laws

The population of a species is initially 2500 and increases by 8% each year. Which function models the population $P(t)$ after t years?

a. $P(t) = 2500 + (1.08)^t$

b. $P(t) = 2500(-7)^t$

c. $P(t) = 2500(8)^t$

d. $P(t) = 2500(1.08)^t$

25 [Foundational] • Target: Solving logarithmic equations

The population of a species is initially 4000 and decreases by 10% each year. Which function models the population $P(t)$ after t years?

a. $P(t) = 4000(11)^t$

b. $P(t) = 4000 + (0.9)^t$

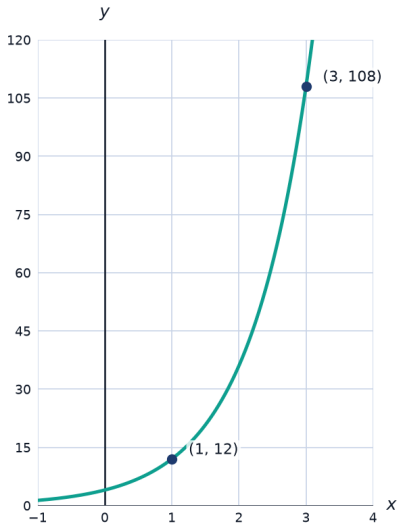
c. $P(t) = 4000(10)^t$

d. $P(t) = 4000(0.9)^t$

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26 [Foundational] • Target: Exponential growth and decay models

The graph of the function $f(x) = a \cdot b^x$ is shown below. If the graph passes through the points $(1, 12)$ and $(3, 108)$, what is the value of a?



a. 2

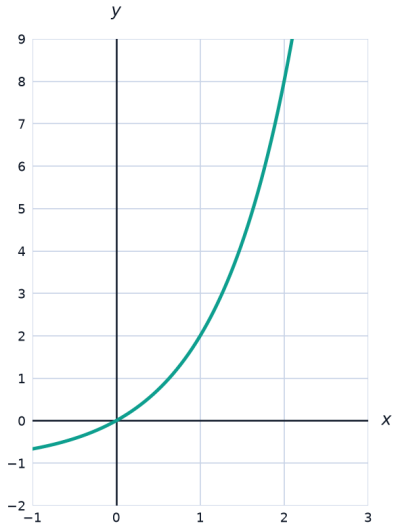
b. 3

c. 4

d. 6

27 [Foundational] • Target: Graphs of exponential functions

Consider the function $f(x) = 3^x - 1$. What is the value of $f(2)$?



a. 5

b. 6

c. 8

d. 9

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28 [Foundational] • Target: Natural logarithms and e

A particle's position x in meters along a straight line is described by the equation $x(t) = e^{2t} - 6e^{t} + 5$, where t is time in seconds. The particle starts at $t = 0$. Find the time $t > 0$ when the particle momentarily stops, and the total distance traveled by the particle from $t = 0$ until it next comes to rest.

a. Time = $\ln 3$ s, Distance = 4 m

b. Time = $\ln 2$ s, Distance = 1 m

c. Time = $\ln 3$ s, Distance = 2 m

d. Time = $\ln 2$ s, Distance = 2 m

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: A well-designed word problem: comparing two investment accounts

If $9^{x+1} = 27^{x-1}$, what is the value of x?

a. 2

b. 3

c. 5

d. 7

30 [Intermediate] • Target: Exponent rules

Which of the following is equivalent to $\log_b\left(\frac{x^5y^3}{z^4}\right)$ for all positive values of x, y, z?

a. $5\log_b(x) \cdot 3\log_b(y) - 4\log_b(z)$

b. $\frac{5\log_b(x) + 3\log_b(y)}{4\log_b(z)}$

c. $5\log_b(x) - 3\log_b(y) - 4\log_b(z)$

d. $5\log_b(x) + 3\log_b(y) - 4\log_b(z)$

31 [Intermediate] • Target: Solving exponential equations

For all positive values of x, which expression is equivalent to $x^{\frac{4}{5}}$?

a. $\frac{x^4}{5}$

b. $\sqrt[5]{x^4}$

c. $\frac{1}{\sqrt[5]{x^4}}$

d. $\sqrt[4]{x^5}$

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32 [Intermediate] • Target: Logarithm definition and evaluation

What is the exact value of $\log_3\left(\frac{1}{3}\right)$?

a. -1

b. 1

c. -3

d. 0

33 [Intermediate] • Target: Logarithm laws

In 2011, the population of Cairo was 9.12 million. Assuming the population increased by 1% each year, what was the approximate population of Cairo in 2022?

a. 10.074154 million

b. 10.123200 million

c. 10.174896 million

d. 10.276645 million

34 [Intermediate] • Target: Solving logarithmic equations

The number of bees in a beehive rose from 2400 to 2520 in one month. How many bees are expected in the next month, assuming the population continues to experience the same monthly percentage increase?

a. 2600

b. 2640

c. 2646

d. 2656

35 [Intermediate] • Target: Exponential growth and decay models

A population starts at 140 and doubles every year. What is the population after 1 years?

a. 140

b. 280

c. 290

d. 420

36 [Intermediate] • Target: Graphs of exponential functions

A population starts at 50 and triples every year. What is the population after 1 years?

a. 200

b. 100

c. 160

d. 150

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37 [Intermediate] • Target: Natural logarithms and e

Given the equation $\log_3(x^2) = 6$, what is the positive value of x?

a. 27

b. 9

c. 81

d. 3

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: A well-designed word problem: comparing two investment accounts • Cross-Lesson: 4.01 Ratios, Rates & Data

Given the equation $2\log_3(x) = \log_3(8x - 15)$, what is the product of all valid solutions for x?

a. 15

b. 8

c. 5

d. 3

39 [Challenging] • Target: Exponent rules • Cross-Lesson: 5.01 Statistics & Probability

Given the equation $\log_2(x) + \log_2(x - 3) = 2$, what is the x?

a. 4

b. 1

c. -1

d. 2

40 [Challenging] • Target: Solving exponential equations • Cross-Lesson: 6.01 Geometry

Given the equation $3^{2x} - 10 \cdot 3^x + 9 = 0$, what is the sum of all real solutions for x?

a. 2

b. 0

c. 1

d. 3

41 [Challenging] • Target: Logarithm definition and evaluation • Cross-Lesson: 7.01 Trigonometry & Additional Topics

Given the equation $2^{2x} - 9 \cdot 2^x + 8 = 0$, what is the larger real solution for x?

a. 3

b. 0

c. 4

d. 2

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42 [Challenging] • Target: Logarithm laws • Cross-Lesson: 1.01 Heart of Algebra

A radioactive sample of 1900 grams has a half-life of 1 year. How much remains after 2 years?

a. 480 g

b. 476 g

c. 950 g

d. 475 g

43 [Challenging] • Target: Solving logarithmic equations • Cross-Lesson: 2.01 Quadratics & Polynomials

Given $\log_{10}(7) \approx p$, what does $\log_{10}(7^4)$ equal, in terms of p?

a. p+4

b. 4p

c. p^4

d. p/4

44 [Challenging] • Target: Exponential growth and decay models • Cross-Lesson: 3.01 Functions & Graphs

A population of 500 grows at 10% per year. Which expression gives its size after t years?

a. $500 + 10t$

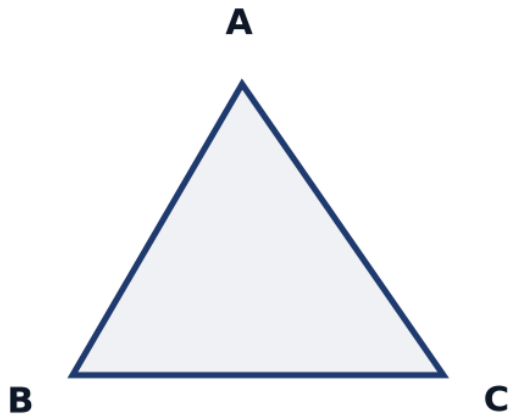
b. $500(1.1)^t$

c. $500(0.1)^t$

d. $500 \cdot 1.1t$

45 [Challenging] • Target: Graphs of exponential functions • Cross-Lesson: 4.01 Ratios, Rates & Data

Two angles of a triangle are 48° and 73°. What is the third angle?



a. 49°

b. 59°

c. 69°

d. 79°

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